

Insert, Select, Where, Update, Delete, Order By, Limit, CRUD Operation

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Connecting PHP with MySQL

config.php

```
<?php
$dbhost = 'localhost';
$dbname = 'class_11';
$dbuser = 'root';
$dbpass = 'root';
try {
    $pdo = new PDO("mysql:host={$dbhost};dbname={$dbname}", $dbuser,
$dbpass);
    $pdo->setAttribute(PDO::ATTR_ERRMODE,
PDO::ERRMODE_EXCEPTION);
} catch( PDOException $exception ) {
    echo "Connection error :" . $exception->getMessage();
}
```

INSERT Query

- The INSERT statement is used to add new rows (records) into a database table.

SQL Syntax:

```
INSERT INTO table_name (column1, column2, column3)  
VALUES (value1, value2, value3);
```

Example:

```
INSERT INTO students (name, email, course)  
VALUES ('Sabbir Ahmed', 'sabbir@example.com', 'Web Development');
```

INSERT Query (Practical Example)

```
<?php
```

```
include "config.php";
```

```
$statement = $pdo->prepare("INSERT INTO students (name,email) VALUES  
(?,?)");
```

```
$statement->execute(['Peter','peter@example.com']);
```

```
echo $pdo->lastInsertId();
```

SELECT Query

- The SELECT statement is used to fetch or retrieve data from one or more tables in a database.
- You can choose to select specific columns to fetch only necessary data, or select all columns using the asterisk (*) wildcard.

SQL Syntax:

-- Retrieve specific columns

```
SELECT column1, column2 FROM table_name;
```

-- Retrieve all columns

```
SELECT * FROM table_name;
```

SELECT Query (Example)

```
$statement = $pdo->prepare("SELECT * FROM students");  
$statement->execute([1]);  
$result = $statement->fetchAll(PDO::FETCH_ASSOC);  
$total = $statement->rowCount();  
foreach ($result as $row) {  
    $name = $row['name'];  
    $email = $row['email'];  
}
```

WHERE Clause

- The WHERE clause is used to filter records in a database table.
- It ensures that operations (like SELECT, UPDATE, or DELETE) apply only to rows that meet specific condition(s).
- Without a WHERE clause, statements like SELECT fetch every single row in the table, while UPDATE and DELETE affect the entire table.

SQL Syntax:

```
SELECT column1, column2  
FROM table_name  
WHERE condition;
```

WHERE Clause (Common Operators)

= (Equal to)

!= or

<> (Not equal to)

>, <, >=, <= (Comparison)

LIKE (Search for a pattern using % wildcard)

IN (Match any value in a specified list)

AND / OR (Combine multiple conditions)

WHERE Clause (Example)

```
$statement = $pdo->prepare("SELECT * FROM students WHERE id=?");  
$statement->execute([1]);  
$result = $statement->fetchAll(PDO::FETCH_ASSOC);  
$total = $statement->rowCount();  
foreach ($result as $row) {  
    $name = $row['name'];  
    $email = $row['email'];  
}
```

UPDATE Query

- The UPDATE statement is used to modify existing records in a table. You specify which columns to change and the new values to assign to them.

Warning: Always use a WHERE clause when executing an UPDATE statement. If you omit the WHERE clause, all rows in the table will be updated!

SQL Syntax:

UPDATE table_name

SET column1 = value1, column2 = value2

WHERE condition;

UPDATE Query (Example)

```
$statement = $pdo->prepare("UPDATE students SET name=?,email=? WHERE  
id=?");  
$statement->execute([$name,$email,$id]);
```

DELETE Query

- The DELETE statement is used to remove existing records from a database table.
- **Warning:** Always use a WHERE clause when performing a DELETE operation. If you omit the WHERE clause, all records in the table will be permanently deleted!

SQL Syntax:

```
DELETE FROM table_name  
WHERE condition;
```

DELETE Query (Example)

```
$statement = $pdo->prepare("DELETE FROM students WHERE id=?");  
$statement->execute([$id]);
```

ORDER BY Clause

- The ORDER BY clause is used to sort the result-set returned by a SELECT query in either ascending or descending order.
- By default, MySQL sorts records in ascending order (ASC).
- To sort records in descending order, you explicitly use the DESC keyword.
- You can sort by numeric values, alphabetical text, or date columns.

SQL Query:

```
SELECT column1, column2  
FROM table_name  
ORDER BY column1 ASC|DESC;
```

ORDER BY Clause (Example)

```
$statement = $pdo->prepare("SELECT * FROM customers ORDER BY name  
ASC");  
$statement->execute();
```

LIMIT Clause

- The LIMIT clause is used to specify the maximum number of records to return from a query.
- It is especially useful on large tables with thousands of rows to improve performance and reduce server load.

SQL Syntax:

-- Retrieve a fixed number of rows

```
SELECT column1, column2 FROM table_name LIMIT number;
```


LIMIT Clause (Example)

```
$statement = $pdo->prepare("SELECT * FROM students LIMIT 2");  
$statement->execute();
```

CRUD

C = Creat

R = Read

U = Update

D = Delete